

# Measuring the External Field: A Cortical-Proxy Content Observable and the Mean-Field Bound on Media-Driven Opinion Change

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## Abstract

Physicists model opinion formation with the Ising model, the same mathematics that describes how magnetic spins in a metal align. In its mean-field version, each person holds one of two opinions, is pulled toward the majority by a coupling  $J$ , and media acts as an external field  $h$  pushing everyone the same way, yet studies choose the value of  $h$  by hand and never measure it. This project asks whether it can be measured from the content itself. Each news article was converted to speech and passed through TRIBE v2, a published AI model from Meta that predicts the brain's response, and the predicted activity in regions linked to emotion was compared with that in regions linked to careful reasoning, giving one number  $X$  per article; no person was scanned, and every value is a model prediction. Four hundred articles in four groups (fear-driven, outrage, clickbait and neutral) were measured twice. The groups differ clearly ( $\eta^2 = 0.107$ ,  $p < 10^{-8}$ ), led by fear-driven articles, and the result repeated on the second run. Each group came from a different source, however, so the difference cannot be credited to writing style alone, and a simple word-counting method sorted manipulative from neutral articles better than this measure, which is therefore offered as a physical quantity rather than a manipulation detector. Writing the field as  $h = \alpha X$ , the strength  $\alpha$  with which content pushes opinion could not be pinned down from the data, so no value is claimed. Instead, the thesis derives a condition that needs no fitted value: in this model, media can flip a population's majority only if  $\alpha \geq h_c(\beta J)/\Delta X$ , where  $h_c$  is the critical field and  $\Delta X$  the spread of the measure. With the measured  $\Delta X = 0.124$ , this requires  $\alpha \geq 4.29$  at  $\beta J = 2$ , a minimum any future media-influence model of this kind must meet.

**Keywords:** opinion dynamics, mean-field Ising model, external field, media influence, brain-response prediction.